Wang Likun 王立坤



Algorithm Engineer · AI Systems & Hardware Acceleration · Shenzhen, China

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Algorithm Engineer working on TPU operator development and toolchain optimization, with an M.Eng. in Information, Production and Systems from Waseda University. Background across computer vision, deep learning, finite element analysis, and bio-inspired structural optimization. Monbukagakusho Honors Scholarship recipient and WCSMO-16 oral presenter.

EXPERIENCE

Algorithm Engineer
Shenzhen, China · Sep 2025 – Present
Shenzhen Suanhai Technology Co., Ltd.

- Develop and optimize operators for a TPU inference stack, focusing on correctness, precision, and performance.
- Improve toolchain workflows for deploying deep learning models on specialized hardware.
- Profile operator-level bottlenecks and validate numerical behavior against reference implementations.

Production Operation Intern
Hefei, China · Aug 2024 – Sep 2024
iFLYTEK Co., Ltd.

- Led core feature testing and optimization for the input-method assistant.
- Analyzed competitors and improved the Quick Phrase and floating-bar experience.

Project Manager Intern
Chengdu, China · Jun 2022 – Jul 2022
Sichuan Huadi Information Technology Co., Ltd.

- Contributed to full-stack development of a mobile homestay-booking application (Android/MyBatis).
- Implemented page layouts, feature modules, debugging, and stability checks.

Product Analyst Intern
China · May 2022 – Jun 2022
Alibaba Entertainment

- Extracted and analyzed user-behavior data with SQL and Python.
- Built Tableau dashboards and wrote reports supporting product strategy.

RESEARCH EXPERIENCE

Fish Vertebrae Loading-Force Prediction
Advisor: Prof. Shintaro Yamasaki · Apr 2024 – Jul 2025
Waseda University · Optimization Lab

- Built a framework combining CT-based 3D reconstruction, COMSOL finite element analysis, topology optimization, and a deep neural network.
- Improved prediction accuracy and computational efficiency by over 20% versus conventional approaches.
- Presented the work orally at WCSMO-16 (Kobe, Japan, May 2025).

Driver Emotion Recognition System
Advisor: Prof. Yalan Ye · Oct 2023 – May 2024
UESTC · School of Computer Science and Engineering

- Developed a real-time facial-expression-based emotion recognition system with a modified YOLOv7 architecture.
- Built a custom dataset, a PyQt5 interface, and a SQLite logging backend.

PUBLICATION & TALK

Wang, L., Yang, L., Sakashita, M., & Yamasaki, S. (2025). A Framework for Predicting Loading Forces Applied on Fish Vertebrae Based on Topology Optimization and Deep Neural Network. Oral presentation at WCSMO-16, Kobe, Japan.

Oral presentation · 16th World Congress of Structural and Multidisciplinary Optimization

EDUCATION

M.Eng. Information, Production and Systems Engineering
Waseda University Fukuoka, Japan · Sep 2023 – Jul 2025

- Advisor: Prof. Shintaro Yamasaki
- Monbukagakusho Honors Scholarship (JASSO)

B.Eng. Computer Science and Technology
University of Electronic Science and Technology of China Chengdu, China · Sep 2020 – Jun 2024

- UESTC–Waseda G-Course joint program
- Class monitor, 2020–2023

Winter Programme, Computer Science and Technology
University of Cambridge Cambridge, UK · Jan 2023 – Feb 2023

AWARDS & HONORS

WCSMO-16 Oral Presentation
World Congress of Structural and Multidisciplinary Optimization · 2025

Monbukagakusho Honors Scholarship
Japan Student Services Organization (JASSO) · 2023

MCM/ICM Honorable Mention
COMAP · 2023

Tencent Game Design Workshop, Silver Award
Tencent Games · 2022

Huawei Cloud "Youth of Promise"
Huawei Cloud · 2022

Innovation & Entrepreneurship Competition, Excellence Award
UESTC · 2022

TECHNICAL SKILLS

PROGRAMMING
Python, SQL, Java, MATLAB

AI / ML
PyTorch, TensorFlow, YOLO, computer vision, data analysis

ENGINEERING
COMSOL, topology optimization, finite element analysis

TOOLS
Git, PyQt5, SQLite, Tableau

LANGUAGES

Chinese	Native
English	Professional
Japanese	Conversational